Beyond the Nearest Actor: A Scenario-Disjoint Audit of Current-Frame ODD-Context Proxies in WOMD

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Abstract

Current-frame TTC screens often compress a multi-agent scene into the SDC and one focal actor, potentially omitting observable roadway and traffic context. We conduct a construct-controlled, scenario-disjoint audit of whether dataset-observable ODD-context proxies provide conditional validity beyond a nearest-clearance SDC–actor baseline. On a processed WOMD v1.3.1 cohort of $18,445$ scenarios (1.67M frames), we pre-lock all development choices and evaluate frozen models once on an internal holdout of $2,804$ scenarios ($255,164$ frames). The binary reference marks frames whose minimum linear-extrapolated swept OBB-TTC between the SDC and any eligible dynamic actor within 70 m is at most 3 s. With scenario-equal weighting and paired scenario-block bootstrap inference, the full context model improves holdout AP from 0.3224 to 0.3370 (ΔAP $= +0.0147$; 95% CI $[+0.0005, +0.0285]$). Family-restricted comparisons localize this increment to non-focal, SDC-relative interaction summaries (ΔAP $= +0.0161$; 95% CI $[+0.0057, +0.0264]$); adding static-roadway or traffic-composition features alone does not improve AP. As a temporal completeness check, the pre-specified closing-pressure-sum anchor remains positively associated with target Peak ($\rho = +0.1914$), AUC ($\rho = +0.2345$), and TET3 ($\rho = +0.0728$) after development-fitted focal-kinematic adjustment, although magnitudes and support vary across features. Because the interaction proxies and TTC target share current-state geometry, these results support modest conditional and convergent surrogate validity within the processed cohort, not causal, orthogonal, crash, or closed-loop safety validity. The audit provides a bounded evaluation practice for testing whether focal scene reductions omit measurable current-frame context.

1. Introduction

Pairwise TTC surrogates characterize instantaneous criticality between two actors, while recorded driving scenes contain additional nearby actors and contextual conditions [3, 4, 8, 15]. Such a reduction is interpretable, but it may omit measurable information carried by other nearby actors, traffic composition, and roadway context [5, 6, 10]. We therefore ask a narrower evaluation question: after controlling for a clean nearest-clearance SDC–actor representation, do dataset-observable current-frame ODD-context proxies improve identification of ego-centric minimum swept OBB-TTC events, and are the resulting proxy associations coherent with temporal target profiles?

Our contributions are threefold:

1. Construct-Controlled Audit Formulation: We define a construct-controlled current-frame audit that compares a nearest-clearance focal baseline with nested ODD-context proxy families for an ego-centric all-nearby-actor TTC target; the target, proxy, and shared-state boundaries are explicit.
2. Sealed Holdout Incremental Finding: We apply a scenario-disjoint one-shot holdout protocol with scenario-equal metrics and paired scenario-block bootstrap inference, finding a supported but modest full-context increment (ΔAP $= +0.0147$, 95% CI $[+0.0005, +0.0285]$). The protocol limits post-hoc tuning, preserves scenario disjointness, and makes the single holdout access auditable.
3. Feature-Family and Temporal Completeness Evidence: Family-restricted comparisons and development-to-holdout feature confirmation provide secondary localization evidence, while residualized scenario-profile analyses test temporal completeness against target Peak, AUC, and TET3. Warp experiments remain excluded and within-dataset vehicle-response KPIs are reported only as supportive development evidence.

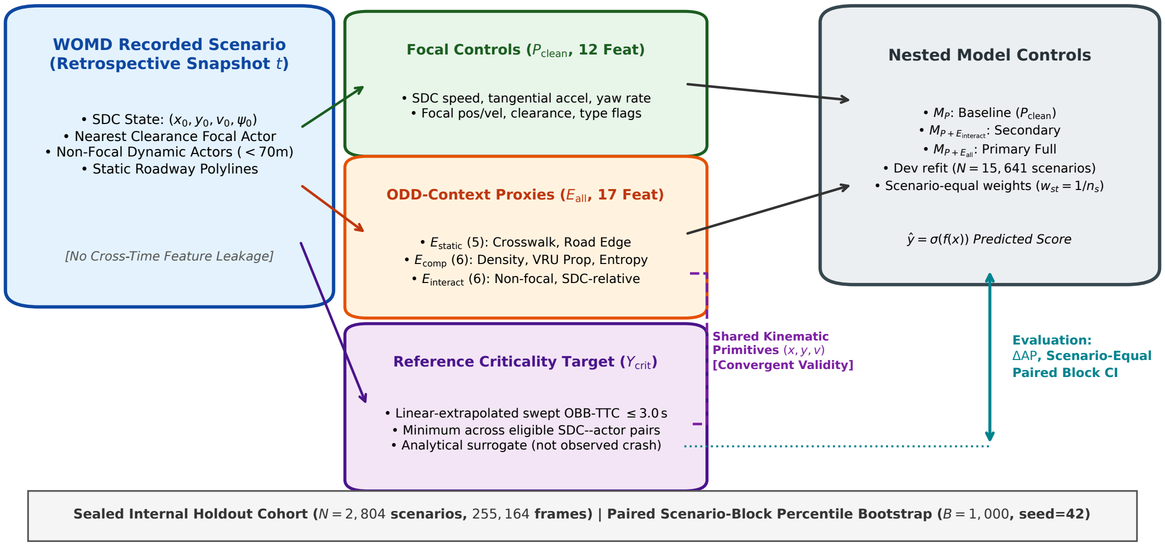

Figure 1. Evaluation Architecture and Construct Boundaries. Overview of the construct-controlled, scenario-disjoint audit protocol. Retrospective snapshot frames extract clean focal physical kinematics (P_{clean}, 12 features) and observable operational-context proxies (E_{all}, 17 features). Nested gradient-boosted models are evaluated against an ego-centric minimum swept SAT OBB-TTC reference target across eligible dynamic actors within 70 m on a sealed one-shot holdout cohort ($N = 255,164$ frames).

2. Related Work

2.1. Safety Surrogates in Autonomous Driving

Time-to-collision (TTC) and its variants (e.g., Modified TTC, Time-Exposed TTC) are foundational surrogates for traffic conflict severity [4, 7, 8, 15]. In automated vehicle development and large-scale public motion forecasting benchmarks (e.g., nuScenes [1], Argoverse [2], WOMD [3]), criticality metrics are widely used for mining scenarios and benchmarking models. SafeShift [12] explores safety-critical distribution shifts under geographical and environmental variations. Puphal et al. [9] propose risk-based filtering in WOMD. Weng et al. [14] emphasize joint multi-agent evaluation. However, benchmark evaluations frequently reduce scenes to focal agent pairs without quantifying what information is omitted.

2.2. ODD-Context Representation and Scene Analysis

Taxonomies for operational design domains (ODD) emphasize static geometry, dynamic actor mix, and environmental conditions [5, 6, 10, 13]. Query-centric and transformer-based motion forecasting architectures (e.g., MTR [11], QCNet [16]) aggregate multi-agent context via attention mechanisms. Together, these studies motivate interaction-aware evaluation; our focus is complementary: we audit whether dataset-observable current-frame ODD-context proxies add predictive information beyond a focal SDC–actor kinematic baseline for an ego-centric TTC target.

3. Problem Formulation and Construct Boundary

3.1. Ego-Centric Target Estimand

Let $\mathcal{A}_{70}(s,t)$ denote eligible dynamic actors with valid current states within 70 m of the SDC in scenario s at frame t. The binary reference label is

$$Y_{s,t}^{(\tau)} = \mathbb{1}\left[\min_{j \in \mathcal{A}_{70}(s,t)} \mathrm{TTC}_{\mathrm{OBB}}(\mathrm{SDC}, j) \leq \tau\right], \quad \tau = 3 \text{ s}. \tag{1}$$

Valid frames with $\mathcal{A}_{70}(s,t) = \emptyset$ are assigned $Y_{s,t}^{(\tau)} = 0$. The TTC is obtained by a constant-velocity swept OBB/SAT extrapolation from the state observed at frame t. Thus, "all-nearby-actor" refers to all eligible SDC–actor pairs, not arbitrary actor–actor pairs. The label is an analytical ego-centric surrogate, not an observed collision, causal risk, or system-safety ground truth.

For feature-level and temporal analyses, we additionally use the continuous score

$$C_{s,t} = \begin{cases} 1, & \mathrm{TTC}_{s,t} = 0, \\ 1 - \mathrm{TTC}_{s,t}/10, & 0 < \mathrm{TTC}_{s,t} < 10 \text{ s}, \\ 0, & \text{otherwise}, \end{cases} \tag{2}$$

where $\mathrm{TTC}_{s,t}$ is the minimum eligible SDC–actor swept OBB-TTC; "otherwise" includes TTC of at least 10 s and the absence of an eligible predicted contact. This bounded score is an analysis surrogate and not a calibrated probability.

3.2. Shared-State Construct Boundary

The interaction proxies summarize non-focal actors relative to the SDC after excluding the nearest-clearance focal actor. Both these proxies and the TTC reference use current-state geometry and velocity. Their association therefore represents shared-state convergent validity conditional on the focal controls; it must not be interpreted as target-independent information, causal influence, or closed-loop safety benefit.

3.3. Temporal Role: Retrospective Snapshot Analysis

Each WOMD training/validation sequence contains 91 samples over 9.0 s at 10 Hz: 10 history samples, one current sample, and 80 future samples relative to the benchmark prediction origin. Our analysis is retrospective. At each evaluated index t, predictors use only the state recorded at that same index, and the TTC target is a linear kinematic extrapolation from that state.

4. Data and Sealed Evaluation Protocol

4.1. Cohort Provenance and Sealed Split

We analyze all $18,445$ WOMD v1.3.1 scenario partitions discovered in the frozen local preprocessing input; the evaluation pipeline applied no further scenario sampling before splitting. The preprocessing artifacts do not retain upstream source-shard coverage as probability-sampling metadata, so this cohort should not be interpreted as representative of the full corpus. A namespace- and seed-fixed SHA-256 assignment partitions the processed cohort by scenario ID into $12,828$ training, $2,813$ internal-validation, and $2,804$ internal-holdout scenarios; all 91 samples from a scenario remain in one split. Development comprises training plus internal validation ($15,641$ scenarios). All model and analysis choices were locked before the single holdout inference pass.

4.2. Feature Taxonomy

P_{clean} contains 12 current-frame focal controls: SDC speed, tangential acceleration, and yaw rate; focal relative longitudinal and lateral position and velocity; focal speed; OBB clearance and center distance; and indicators for vehicle and vulnerable-road-user focal types. The focal actor is selected by current OBB clearance independently of the TTC outcome. The 17 ODD-context proxies comprise five static-roadway features (distances to crosswalk, stop sign, speed bump, and road edge, plus local lane-heading dispersion), six traffic-composition features (actor counts within 30/50/70 m, vehicle and VRU proportions within 70 m, and actor-type entropy), and six non-focal SDC-relative interaction features (nearest distance, mean speed, speed standard deviation, heading dispersion, and maximum/summed closing pressure).

4.3. Model Setup and Scenario-Equal Weighting

All nested models use HistGradientBoostingClassifier(max_iter=100, max_depth=6, random_state=42) under scikit-learn 1.7.1; unspecified parameters retain that version's defaults. Development-stage choices were made using the training/internal-validation partition and then locked. Each final frozen model was refit on all $15,641$ development scenarios using scenario-equal frame weights, and the holdout pass contained zero fit calls. For scenario s with n_s valid frames, each frame receives $w_{s,t} = 1/n_s$, so every scenario contributes equal total weight to AP, AUROC, and Brier score.

4.4. Statistical Inference Protocol

We use 1,000 paired scenario-block bootstrap replicates with seed 42. Each replicate samples holdout scenario IDs with replacement, retains every frame of each sampled scenario, and evaluates both members of a contrast on the identical draw. Within each occurrence, frame weights again sum to one per sampled scenario. We report percentile 95% confidence intervals for paired metric differences; we do not interpret bootstrap sign-tail counts as calibrated p-values.

4.5. Individual Conditional Feature Effect Definition

For feature-level confirmation, we use continuous frame severity $C_{s,t}$ rather than the binary threshold label. On development data only, separate Ridge models with $\alpha = 1$ regress each candidate proxy E_i and C on P_{clean}, using scenario-equal weights. Frozen nuisance models produce holdout residuals $r(E_i)$ and $r(C)$; the reported conditional effect is Spearman's $\rho[r(E_i), r(C)]$. Confidence intervals resample scenarios as blocks. The 13 candidates were frozen before holdout access.

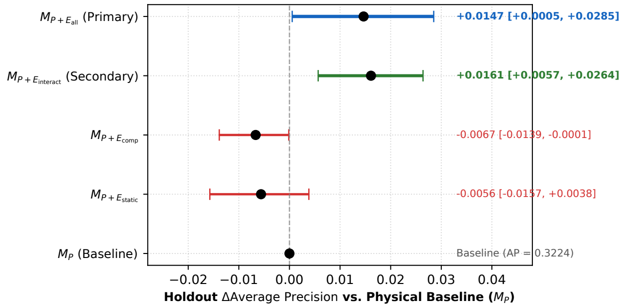

Figure 2. Forest Plot of Holdout Model Contrasts. Paired ΔAP and 95% scenario-block bootstrap confidence intervals relative to physical baseline M_P. The full context model $M_{P+E_{\text{all}}}$ (primary) and interaction model $M_{P+E_{\text{interact}}}$ (secondary) achieve strictly positive lower bounds.

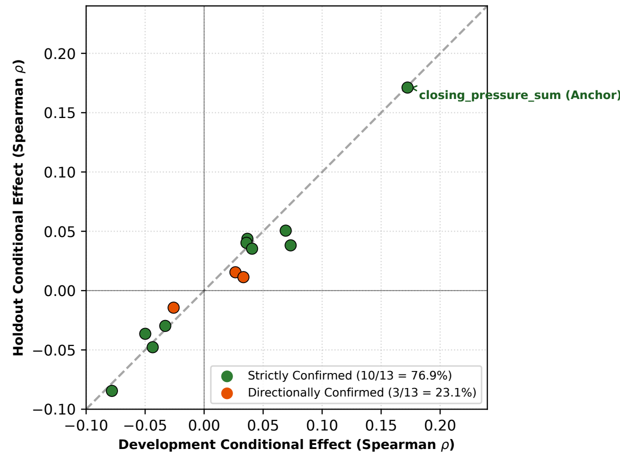

Figure 3. Feature Effect Concordance (Development vs. Holdout). Spearman rank correlation conditional effects across 13 candidate features showing 100% sign concordance and 76.9% strict confirmation.

5. Current-Frame Incremental Validity

5.1. Primary Holdout Evaluation

Table 1 presents the sealed holdout evaluation across nested model architectures on $N = 255{,}164$ frames.

The primary contrast supports a positive but modest conditional increment: AP increases from 0.3224 to 0.3370 (ΔAP $= +0.0147$, 95% CI $[+0.0005, +0.0285]$). The lower bound is close to zero, so the result should not be described as a large or practically universal improvement. The context-only model is substantially weaker than the focal baseline (0.1141 vs. 0.3224), indicating that the proxies are useful conditionally rather than as a standalone replacement for focal kinematics.

5.2. Individual Feature Replication

Across the 13 pre-frozen candidate features evaluated in development, 10 out of 13 (76.9%) achieved strict holdout confirmation with 95% bootstrap confidence intervals strictly excluding zero in the expected direction. Furthermore, 13 out of 13 (100%) demonstrated exact sign concordance between development and holdout splits, as illustrated in Figure 3. Complete numerical breakdowns for all candidate features are provided in the Supplementary Material.

6. Feature-Family and Temporal Completeness Evidence

6.1. Interaction-Family Localization

Family-restricted comparisons localize the positive increment to the non-focal interaction family: $M_{P+E_{\text{interact}}}$ yields ΔAP $= +0.0161$ (95% CI $[+0.0057, +0.0264]$). Adding static features alone does not improve AP (ΔAP $= -0.0056$), while adding composition features yields a small decrease (ΔAP $= -0.0067$). These nested comparisons are consistent with interaction summaries accounting for the observed increment within the evaluated model family; they do not identify a causal mechanism or establish that map and composition context are generally irrelevant.

6.2. Scenario Profile Temporal Completeness

We use scenario profiles as a temporal completeness check, not as the primary estimand. For each proxy i, we compute its scenario mean $\bar{E}_{s,i} = T_s^{-1} \sum_t E_{s,t,i}$. From continuous frame severity $C_{s,t}$ and TTC, we form three target profiles: $D_s^{\text{Peak}} = \max_t C_{s,t}$, $D_s^{\text{AUC}} = \sum_t C_{s,t} \Delta t$, and $D_s^{\text{TET3}} = \sum_t \mathbb{1}[\text{TTC}_{s,t} \leq 3\text{ s}] \Delta t$. Separate Ridge models with $\alpha = 1$, fitted on development scenario means of P_{clean}, residualize $\bar{E}_{s,i}$ and each target profile. We report holdout Spearman correlations between the corresponding residuals with scenario-level bootstrap intervals.

The pre-specified closing-pressure-sum anchor is positive at the frame level ($\rho = +0.1712$) and for target Peak ($+0.1914$, 95% CI $[+0.1549, +0.2300]$), target AUC ($+0.2345$, $[+0.2040, +0.2677]$), and target TET3 ($+0.0728$, $[+0.0405, +0.1086]$). These results support anchor-level temporal completeness, but the heterogeneous magnitudes—particularly the smaller TET3 association—and the mixed 17-feature taxonomy warrant a feature-dependent, not uniformly strong, conclusion. Across the full 17-feature taxonomy, 8 features are classified as CROSS_LEVEL_STABLE, 6 as UNSUPPORTED_BOTH, 1 as FRAME_LOCAL, 1 as DISCORDANT_SIGN, and 1 as TEMPO-

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Table 1. Primary Nested Model Evaluation on Sealed Holdout Cohort ($N = 255,164$ frames, $2,804$ scenarios). Models trained on all $15,641$ development scenarios and evaluated on the sealed holdout. Primary contrast is $M_{P+E_{\text{all}}}$ vs. M_P. Confidence intervals are pre-specified 95% paired scenario-block percentile bootstrap intervals ($B = 1,000$).

Model	Feature Set	Dim	AP	AUROC	Brier	ΔAP vs. M_P	95% CI
M_P (Physical Baseline)	P_{clean}	12	0.3224	0.8022	0.0451	Baseline	—
M_E (Context Only)	E_{all}	17	0.1141	0.7113	0.0516	-0.2082	[-0.2241, -0.1925]
$M_{P+E_{\text{static}}}$	$P_{\text{clean}} + E_{\text{static}}$	17	0.3167	0.8046	0.0452	-0.0056	[-0.0157, +0.0038]
$M_{P+E_{\text{comp}}}$	$P_{\text{clean}} + E_{\text{comp}}$	18	0.3157	0.8078	0.0453	-0.0067	[-0.0139, -0.0001]
$M_{P+E_{\text{interact}}}$ (Secondary Core)	$P_{\text{clean}} + E_{\text{interact}}$	18	0.3385	0.8347	0.0444	+0.0161	[+0.0057, +0.0264]
$M_{P+E_{\text{all}}}$ (Primary Full)	$P_{\text{clean}} + E_{\text{all}}$	29	0.3370	0.8399	0.0444	+0.0147	[+0.0005, +0.0285]

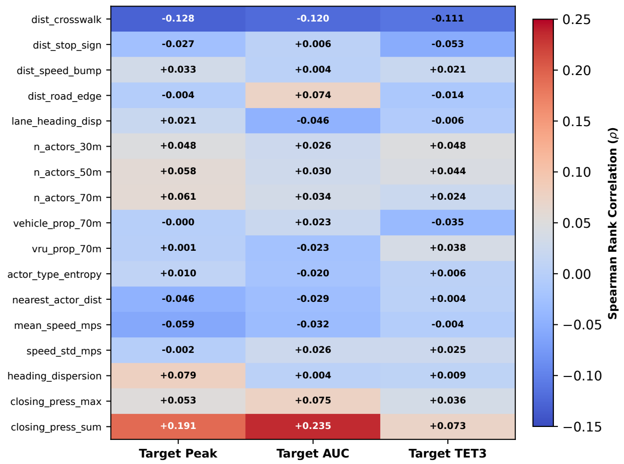

Figure 4. Temporal Target-Profile Completeness. Holdout residual Spearman associations between each scenario-mean ODD-context proxy and target Peak, AUC, and TET3 after development-fitted adjustment for scenario-mean focal kinematics ($N = 2,804$ scenarios). Column labels refer to target profiles, not transformations of the proxy itself.

RAL_EXPOSURE_SPECIFIC.

7. Discussion, Limitations, and Intended Use

7.1. Threshold Horizon Sensitivity

Sensitivity results establish a horizon boundary rather than a behavioral cause. The context increment is positive at 3 s (ΔAP $= +0.0147$), larger at 5 s (ΔAP $= +0.0660$), and negative in the rarer 2 s tail (ΔAP $= -0.0442$). The available analysis does not determine whether this variation arises from prevalence, calibration, model capacity, or a change in the underlying event regime; we therefore avoid attributing the 2 s result to post-encroachment reaction behavior.

7.2. Supportive Development KPI Construct Validity

Within-dataset vehicle-response KPIs provide supportive development-only convergent evidence. For example, the continuous TTC-based severity score $C_{s,t}$ is modestly associated with SDC hard-deceleration p95 (Spearman $\rho = +0.1347$; Cohen's $d = +0.3451$). Other KPIs are weaker or directionally mixed, including the DRAC summaries. Because these KPIs are computed from the same recorded sequences and share kinematic state, they are neither external validation nor evidence of crash avoidance.

7.3. Limitations and Excluded Explorations

This study is a retrospective audit of analytical surrogates on a processed WOMD cohort. The TTC target assumes constant current velocity and OBB geometry and is not an observed crash label. Predictors and target share current-state primitives, so the evidence is convergent rather than orthogonal or causal. The processed $18,445$-scenario cohort is not the complete WOMD corpus, and the internal holdout does not establish cross-dataset, geographic, or population validity. Recorded trajectories cannot evaluate closed-loop policy response or the safety performance of any deployed driver. Threshold sensitivity shows that the primary conclusion does not extend unchanged to every TTC horizon. Warp experiments were inconclusive and are excluded; within-dataset KPI analyses remain supportive and development-only.

8. Conclusion

We presented a scenario-disjoint audit of whether current-frame ODD-context proxies add conditional validity beyond a nearest-clearance focal-kinematic baseline for an ego-centric all-nearby-actor TTC surrogate. On a one-shot holdout, the full proxy set produced a supported but modest AP increment (ΔAP $= +0.0147$, 95% CI $[+0.0005, +0.0285]$). Family-restricted comparisons localize the increment to non-focal SDC-relative interaction summaries, whereas static and composition families do not improve AP when added alone. The pre-specified interaction anchor also remains positive across residualized target Peak, AUC, and TET3 profiles, with feature-dependent support across the full taxonomy. Within the processed cohort, these findings provide a bounded evaluation result: focal scene reductions can omit measurable current-frame context. They do not establish population-wide WOMD validity, causal risk, crash prediction, or closed-loop system safety.

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